Chapter 13

High Frequency Trading and Black Box Models

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INTRODUCTION

The growth in active investing has seen the rise of high frequency trading. Since the credit crisis and the subsequent flurry of regulations with respect to derivatives, no area of finance has been under such scrutiny as high frequency trading. High frequency trading is characterized by an enhanced turnover of capital in response to market dynamics. This usually results in enhanced trading activity coupled with smaller gains per trade. In contrast with traditional investing and trading, the holding period and thus the investment horizon is orders of magnitude smaller.

High frequency trading strategies themselves have found preponderance amongst money managers with most opting to close these strategies out daily (Aldridge, 2009). Closing out positions overnight provides three key advantages to a money manager:

- Through the proliferation of trading and execution venues, trading is now virtually a 24-hour a day activity. Thus overnight positions, given the slivers they are trying to capitalize on, could become extremely risky. Closing flat at the end of trading in one's main market removes this risk.
- Capital is not committed beyond the course of close of business, enabling transparent accounting.
- Typically, overnight positions taken out on margin are paid for at the overnight carry rate which is usually slightly above LIBOR. Overnight LIBOR volatility and the chances of hyperinflation can quickly deteriorate with overnight positions becoming increasingly expensive and diminishing any profits accrued. Closing out avoids overnight carry risk, providing substantial savings in tight lending and high-interest regimes.

Furthermore, it has been argued that high frequency trading contributes to market efficiency. There is added liquidity, reduced trading costs and

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